2.6 Other metrics

- The best approach → measure CPU time on Real Programs
- MIPS, MFLOPS
  Million Instructions Per Second
  Million Floating-Point Operations Per Second

$$MIPS = \frac{IC}{\text{Execution time} \times 10^6} = \frac{IC}{IC \times CPI \times \text{Clock cycle time} \times 10^6} =$$

$$= \boxed{\frac{\frac{\text{CPU time}}{\text{Clock rate}}}{CPI \times 10^6}} \qquad \text{Execution time} = \frac{IC}{MIPS \times 10^6}$$

MIPS is a speed metric!

Proofs for MIPS inconsistency

① MIPS depend on the instructions set => not fair to compare 2 machines w/ different instruction sets

② MIPS may vary from program to program on the machine

③ Sometimes, it is possible that MIPS varies inversely with performance

$$N_1 = m_1 \times B^{E_1}$$
$$N_2 = m_2 \times B^{E_2}$$
$$N_1 \times N_2 = (m_1 \times m_2) \times B^{E_1 + E_2}$$

Example We consider a machine with the following instruction mix when running a program.

Op	Freq	Clock cycle count
ALU op	43%	1
Loads	21%	2
Stores	12%	2

- We consider a compiler optimization => 50% reduction of ALU Ops
- We consider that the clock cycle time of 20ns

$MIPS_{\dots} = \frac{\text{Clock rate}}{\dots}$

Stores	12%	2
Branches	24%	2

$$MIPS_{unoptimized} = \frac{Clock\ rate}{CPI_u \times 10^6}$$

$$\begin{array}{r} 1.14 \\ 0.43 \\ \hline 1.57 \end{array}$$

$$Clock\ rate = \frac{1}{20 \cdot 10^{-9}\ s} = \frac{10^9}{20}\ Hz = \frac{1000}{20} \cdot 10^6\ Hz = 50\ MHz$$

$$CPI_u = 0.43 \times 1 + 0.57 \times 2 = 1.57\ c.c.$$

$$MIPS_{unoptimized} = \frac{50 \cdot 10^6\ Hz}{1.57 \times 10^6} = \boxed{31.85}$$

$$CPU_{time\ unoptimized} = IC_u \times 1.57 \times 20 \cdot 10^{-9}\ s = \boxed{IC_u \times 31.4\ us}$$

$$CPI_o = \frac{\frac{0.43}{2} \times 1 + 0.57 \times 2}{1 - \frac{0.43}{2}} = \frac{1.355}{0.785} = 1.73\ c.c.$$

$$MIPS_{optimized} = \frac{50 \cdot 10^6\ Hz}{1.73 \cdot 10^6} = \boxed{28.9}$$ ✗

$$CPU_{time\ optimized} = IC_o \times CPI_{optimized} \times 20\ us =$$

$$= \left(1 - \frac{0.43}{2}\right) \times IC_u \times 1.73 \times 20\ us =$$

$$= 0.785 \times IC_u \times 1.73 \times 20\ us =$$

$$= \boxed{IC_u \times 27.161\ us}$$ ✓

- This is native MIPS!
- Relative MIPS is more fair $\quad Relative\ MIPS = \frac{Execution\ time_{Reference}}{Execution\ time_{unrated}} \times MIPS_{reference}$
- 1-MIPS machine → VAX 11/780

• Conclusion: Use MIPS with extreme caution in both its forms

$$\text{MFLOPS} = \frac{\text{No of FP instructions}}{\text{Execution time} \times 10^6}$$

- More fair than MIPS
- MFLOPS is also a speed-type metric

What is wrong with MFLOPS?

① Relevant only when the computer operates in FP

② Different FP instruction sets, e.g., some machines have exp, sin, cos, $\sqrt{}$ as instructions, and some do not.

③ Different machines have different fixed / floating point op. ratios, different simple vs. complex FP op ratios.

Mc Mahon → Lawrence Livermore Laboratory

→ Livermore Loops

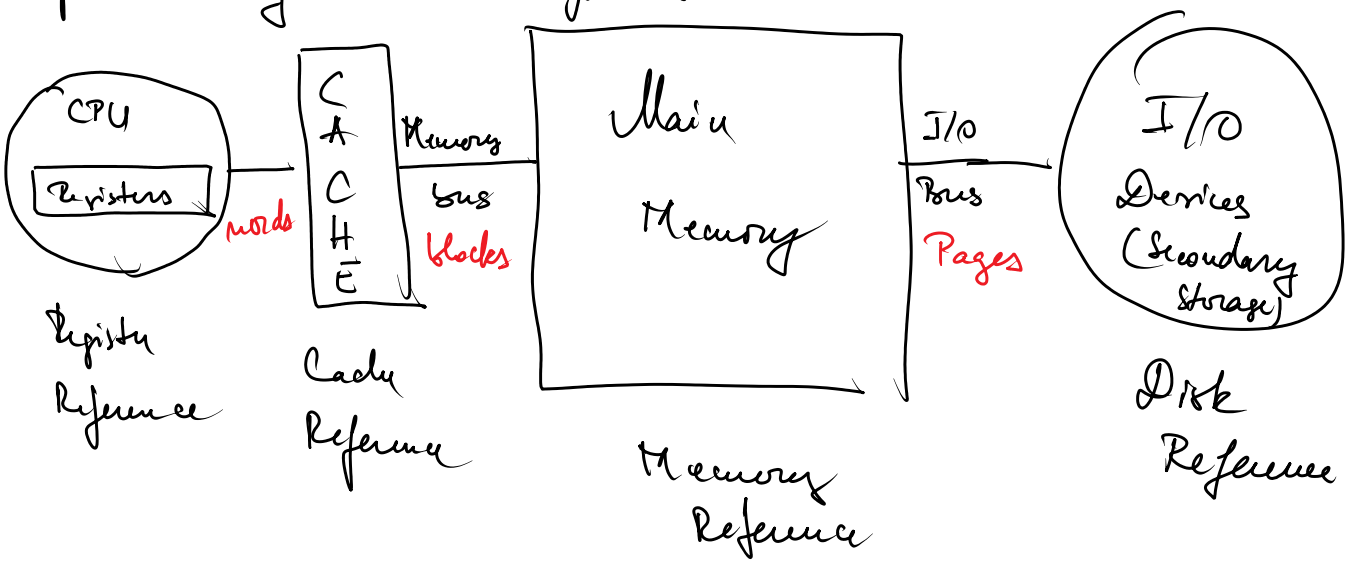
No of FP operations → Normalized FP operations

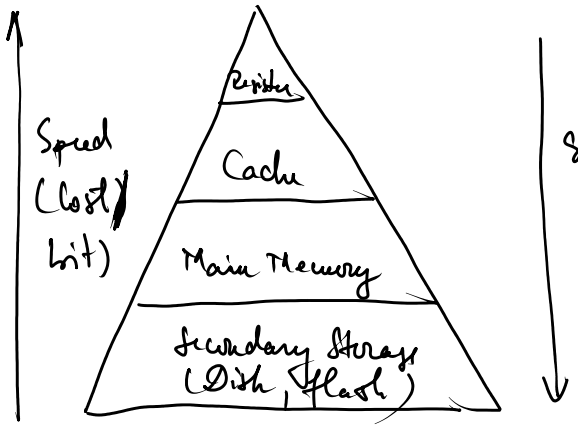
Real FP operations	Normalized FP operations
+, −, comp, ×	1
Divide, sqrt	4
Exp, Sin	8

Chapter 3 Memory hierarchies

3.1 Introduction

- Computer memory is not monolythical





Driving principle → Locality

A processor spends 90% of the time on 10% of the code

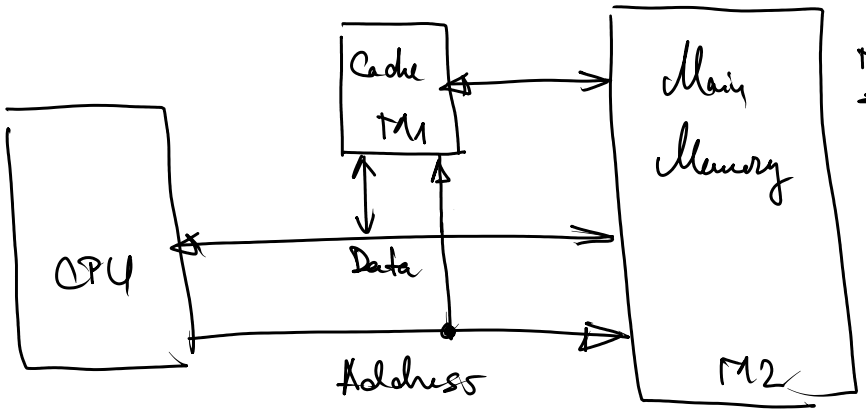
1 Address word = 32 bits

Addressable unit → 1 byte = 1B

Size = 2^{32} B = $2^2 \cdot 2^{30}$ B = 4GiB

Memory
- bit
- bytes – 1 Byte = 1B = 8 bits
- 1 word = 2^p B – CPU register
- 1 block = 2^n words
- 1 page = 2^k blocks

3.2 Caches



M_1, M_2 – divided into blocks

M_2 – divided into pages

3.2.1 Direct mapping

M_2 is divided in 2^u blocks

$M_2(0), M_2(1), \dots, M_2(j), \dots M_2(2^u-1)$; $S_{M_2} = 2^u$ blocks

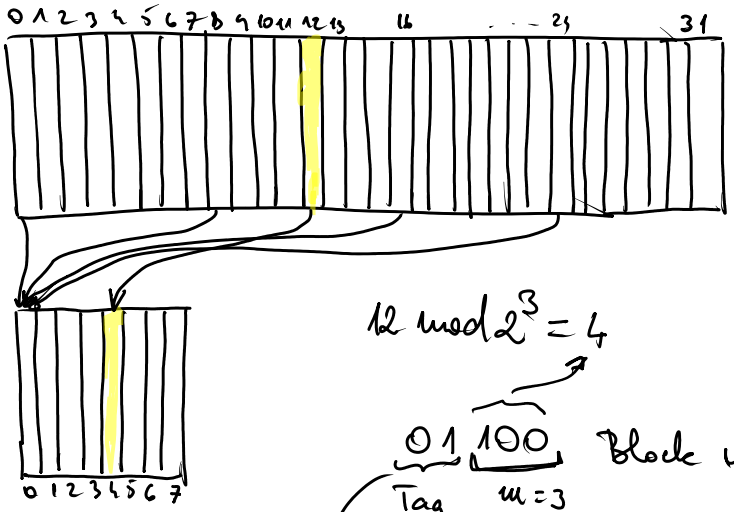
Each block is 2^w words, and each word is 2^b bytes

M_1 is divided in 2^m blocks $\quad m < u$

$M_1(0), M_1(1), \dots, M_1(i), \dots, M_1(2^m-1)$ $\quad Size_{M_1} = 2^m$ blocks

$$i \equiv j \bmod 2^m$$

Example $u = 5$, $m = 3$



$12 \bmod 2^3 = 4$

01 100 Block number

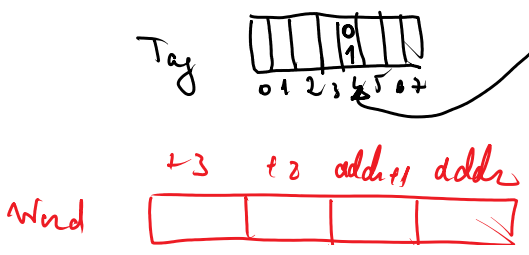
Tag $m = 3$

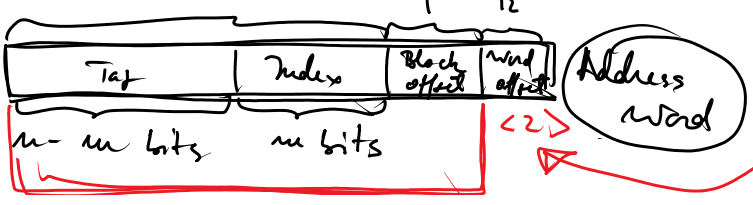
1 word = 32 bits = 2^2

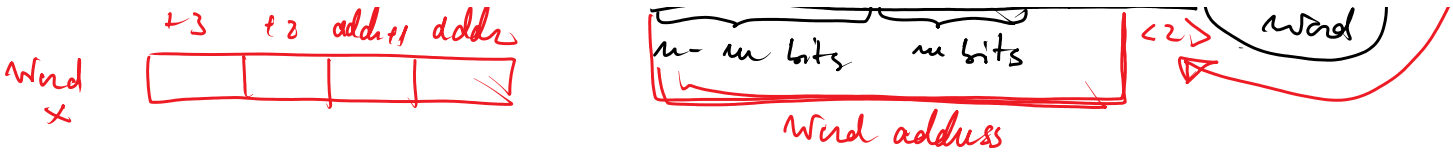
1 byte – addressable

1 word = $\frac{2^5}{2^3} = 2^2$

1 word = 2^2 bytes






t3
t2
addr+1
addr
Word
x
m bits
m bits
<2>
Word
Word address